FIXED-POINT ITERATION AND FEIGENBAUM'S DELTA

From fixed points to the edge of chaos

An A-level numerical method leads from the plastic number to a universal scaling constant.

1 A root becomes a fixed point

$x^3 = x + 1 \quad \Longrightarrow \quad x_{n+1} = g(x_n) = \sqrt[3]{x_n + 1}$. The limit is the **plastic number**:

$$\rho = 1.3247179572\ldots, \qquad g'(\rho) = \frac{1}{3\rho^2} \approx 0.190.$$

Since $g'(\rho)$ is between 0 & 1, nearby iterates move toward ρ.

The local slope decides

At a fixed point $a = g(a)$, let $e_n = x_n - a$.

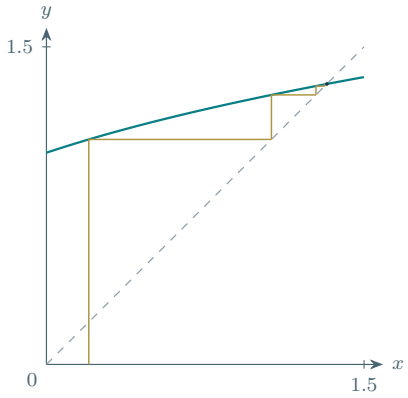
$$e_{n+1} = g(a + e_n) - g(a) \approx g'(a) e_n.$$

Small errors shrink when $|g'(a)| < 1$ and grow when $|g'(a)| > 1$. At $|g'(a)| = 1$, this test is inconclusive.

2 Same stepping rule, different destinations

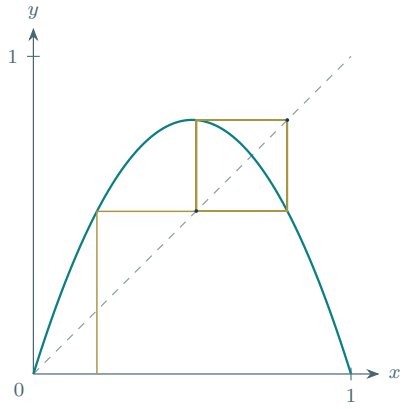
Move vertically to $y = g(x)$ and horizontally to $y = x$, then repeat. Here $x_0 = 0.2$.

Plastic number
$g(x) = \sqrt[3]{x + 1}$



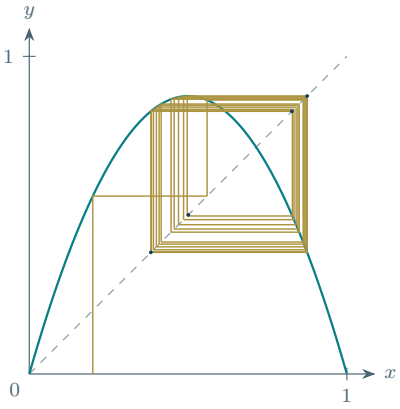
One attracting value: ρ.

An attracting 2-cycle
$g(x) = 3.2x(1 - x)$



Two values repeat.

An attracting 4-cycle
$g(x) = 3.5x(1 - x)$



Four values repeat.

Teal: map dashed: $y = x$ gold: cobweb navy: limiting values on the diagonal

3 Swap the map, keep the iteration

$$g_r(x) = rx(1 - x), \qquad x_* = 1 - \frac{1}{r} \quad (r > 1), \qquad g_r'(x_*) = 2 - r.$$

For $0 \le r \le 4$, the map keeps $[0, 1]$ inside itself. For $1 < r < 3$, the nonzero fixed point attracts. At $r = 3$, its slope reaches -1; an attracting 2-cycle appears just beyond this value. A q-cycle returns after q steps: $g_r^{\circ q}(a) = a$, where $\circ q$ denotes q-fold composition. The stable periods then double: $1 \to 2 \to 4 \to 8 \to 16 \to \cdots$.

4 Where do the doublings occur?

Let r_m be the mth doubling threshold.

m	period change	r_m
1	$1 \to 2$	3.000000000
2	$2 \to 4$	3.449489743
3	$4 \to 8$	3.544090360
4	$8 \to 16$	3.564407266
5	$16 \to 32$	3.568759420

The first two are exact: $r_1 = 3$ and $r_2 = 1 + \sqrt{6}$. The successive parameter intervals become smaller.

5 A constant from shrinking gaps

$$\boxed{\delta = \lim_{m \to \infty} \frac{r_m - r_{m-1}}{r_{m+1} - r_m}}$$

$$\boxed{\delta = 4.6692016091\ldots}$$

Successive ratios from the table approach:

$$4.7514, \quad 4.6563, \quad 4.6682, \quad \ldots$$

The gaps shrink asymptotically by

$$\frac{1}{\delta} \approx 0.2142.$$

Thus each new doubling occupies about 21.4% of the preceding parameter interval.

6 The edge of chaos

The cascade accumulates at $r_\infty \approx 3.569945672$. Beyond this boundary lie chaotic bands and periodic windows.

Many smooth, one-peaked maps with a quadratic maximum share the same δ along their period-doubling cascades.
An iteration used to find one number has led us to a universal scale for the loss of stability.

Plastic number: V. Iliopoulos (2015), arXiv:1407.2091. Dynamics: R. M. May (1976), doi:10.1038/261459a0. Scaling: M. J. Feigenbaum (1978), doi:10.1007/BF01020332.